

## 7. Learning via Uniform Convergence.. and Hoeffding ineq

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### §4.1[SS] Uniform Convergence

A **sufficient** condition for PAC learning  
(**necessary** in some cases, cf. Thm. 6.7, though modern perspective is that uniform convergence is often asking for too much)

**Def** A training set  $S$  is an  **$\varepsilon$ -representative sample** (with respect to set  $Z = X \times Y$ , set  $\mathcal{H}$ , loss  $\ell$ , distr.  $\mathcal{D}$ ) if **hard if  $\mathcal{H}$  big, easier if  $\mathcal{H}$  small**

$$(\forall h \in \mathcal{H}) \quad |\hat{L}_S(h) - L_D(h)| \leq \varepsilon \quad \rightarrow \text{really, 2 req: } \begin{matrix} \hat{L} - L \leq \varepsilon \\ L - \hat{L} \leq \varepsilon \end{matrix}$$

hence "uniform" flavor, eg.  $\|\hat{L}_S - L_D\|_\infty \leq \varepsilon$

Such a set is quite nice:

**Lemma 4.2** If  $S$  is  $\varepsilon/2$ -representative then if  $h_S \in \text{ERM}_{\mathcal{H}}(S)$   
then  $L_D(h_S) \leq \left( \min_{h \in \mathcal{H}} L_D(h) \right) + \varepsilon$  i.e.  $\in \arg\min_{h \in \mathcal{H}} \hat{L}_S(h)$

nothing random

exactly what we want for PAC learning

**proof:**

$$\text{let } h \in \arg\min_{h \in \mathcal{H}} L_D(h)$$

$$\begin{aligned} L_D(h_S) &\leq \underbrace{\hat{L}_S(h_S)}_{\text{by } \varepsilon/2\text{-repr.}} + \varepsilon/2 \leq \hat{L}_S(h) + \varepsilon/2 \quad [\text{since } h_S \text{ minimizes } \hat{L}_S] \\ &\leq L_D(h) + \varepsilon/2 + \varepsilon/2 \quad [\text{by } \varepsilon/2\text{-repr. again}] \\ &= \left( \min_{h \in \mathcal{H}} L_D(h) \right) + \varepsilon \quad \square \end{aligned}$$

Building on this concept...

**Def** A hypothesis class  $\mathcal{H}$  has the **uniform convergence property**

(w.r. respect to  $Z, \ell$ ) if  $\exists m_{\mathcal{H}}^{uc} : (0, 1)^2 \rightarrow \mathbb{N}$  s.t.

$\forall \varepsilon, \delta \in (0, 1), \quad \forall \text{ prob. distr. } \mathcal{D} \text{ over } Z = X \times Y,$

if  $S$  has  $m \geq m_{\mathcal{H}}^{uc}(\varepsilon, \delta)$  iid samples, then

$S$  is  $\varepsilon$ -representative with probability  $\geq 1 - \delta$

Uniform over  
 $h \in \mathcal{H}$  and  
all possible  
 $\mathcal{D}$

**Corollary** If  $\mathcal{H}$  has the **uniform convergence property** with  $m_{\mathcal{H}}^{uc}$ ,

then ①  $\mathcal{H}$  is **agnostic PAC learnable**

②  $m_{\mathcal{H}}(\varepsilon, \delta) \leq m_{\mathcal{H}}^{uc}(\varepsilon/2, \delta)$

③ **ERM** $_{\mathcal{H}}$  is a valid algorithm to learn  $\mathcal{H}$

(proof is immediate)

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§4.2 Finite classes ( $|H| < \infty$ ) have the uniform conv. property

Fix  $\varepsilon, \delta$ . We want  $S$  to be  $\varepsilon$ -representative w.p.  $\geq 1 - \delta$ ,

i.e.  $\mathbb{D}^m(\{S : \forall h \in H, |\hat{L}_S(h) - L_D(h)| \leq \varepsilon\}) \geq 1 - \delta$

or  $\mathbb{D}^m(\{S : \exists h \in H \text{ st. " } \dots \text{ " } > \varepsilon\}) < \delta$

Subtle!

event  $E$

$$E = \bigcup_{h \in H} \{S : |\hat{L}_S(h) - L_D(h)| > \varepsilon\}$$

event  $E_h$

Same trick we did last Subtle! lecture

So via union bound,  $\mathbb{D}^m(E) \leq \sum_{h \in H} \mathbb{D}^m(E_h)$

Rmk: Union bound aka Boole's Ineq.

So look at  $\mathbb{D}^m(E_h)$

(Before, in realizable case,  $\hat{L}_S(h) = 0$  and we could break  $\mathbb{D}^m(E_h)$  into  $\prod_{i=1}^m \mathbb{D}(\dots)$  via independence ...

this trick no longer applies )

Since  $\hat{L}_S(h) = \frac{1}{m} \sum_{i=1}^m l(h, z_i)$  and  $z_i \sim D$  iid, we have  $\mathbb{E} \underbrace{\hat{L}_S(h)}_{\text{r.v.}} = \underbrace{L_D(h)}_{\text{its mean}}$

So today's lecture is statistics: how likely is a r.v. to deviate significantly from its mean?

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= Statistics Interlude =

**BASICS:**  $E[Z] := \int z \cdot D(dz) = \int z dD$ ,  $\sigma^2 = E[Z^2] - E[Z]^2$

**Independence:**  $\theta_1 \perp \theta_2 \Rightarrow E[\theta_1 \theta_2] = E[\theta_1]E[\theta_2]$ ,  $P[\theta_1 = a, \theta_2 = b] = P[\theta_1 = a]P[\theta_2 = b]$   
 $Var(\theta_1 + \theta_2) = Var(\theta_1) + Var(\theta_2)$

**In-class quiz:** test  $\nearrow$  see which students think are always true,  
true if independent, and not even true if independent  
 $\nearrow$  ex:  $E[\theta_1 + \theta_2] = E[\theta_1] + E[\theta_2]$   
 $Var(3\theta) = 9 \cdot Var(\theta)$

1) as  $m \rightarrow \infty$ , the  $\nearrow$  <sup>in probability (weak law)</sup> <sup>or almost sure (strong law)</sup> **law of large numbers** says  
Fixed h  $\hat{L}_{S(m)}(h) \rightarrow L_D(h)$ , but there's no **rate**

2) **Central Limit Thm** says  $\sqrt{m} \hat{L}_S(h) \sim \text{Normal}$  as  $m \rightarrow \infty$   
 (basic CLT has no rate, but Berry-Esseen CLT does)  
 Could we use facts about normal distr.?  
 No, not good bounds: see **Vershynin book**

eg  $l(h, z_i)$   
 $\downarrow$   
 Specifically, if  $Var(\theta_i) = \sigma$   
 $\frac{\hat{L}_S(h) - L_D(h)}{\sigma/\sqrt{m}} \xrightarrow{\text{approx.}} N(0,1)$   
 $\nearrow$  good approx if  $m \geq 30$  if  $\sigma$  known  
 If  $\sigma$  unknown,  
 $\frac{\hat{L}_S - L_D}{\hat{\sigma}/\sqrt{m}} \sim t(m-1)$  if  $l(h, z_i)$  is normal  
 $\nearrow$  Student t

3) What about **Chebyshev's Inequality**?

Recall **Markov's Ineq**: if  $Z \geq 0$  is a r.v. then  
 $P[Z \geq a] \leq \frac{1}{a} \cdot E[Z]$

**Proof:**  $E[Z] = \int z d\mu = \int_{z \geq a} z d\mu + \int_{z < a} z d\mu$   
 $\geq \int_{z \geq a} a d\mu + 0$   
 $= a \cdot P[Z \geq a]$

Let  $\theta$  be a r.v.  
 w/ variance  $\sigma^2$ , then Chebyshev's Ineq. says  
 $\forall \epsilon > 0$ ,  $P[|\theta - E[\theta]| > \epsilon] \leq \frac{\sigma^2}{\epsilon^2}$

**Proof:**  
 apply Markov  
 to  $|\theta - E[\theta]|^2$

Let's see if we can apply to our situation

Define  $\theta_i = l(h, z_i)$ , a r.v. w/ mean  $L_D(h)$

So  $\theta = \hat{L}_S(h) = \frac{1}{m} \sum \theta_i$

What is  $Var(\theta)$ ? Suppose  $Var(\theta_i) = \sigma^2$

$Var(\frac{1}{m} \sum \theta_i) = \frac{1}{m^2} Var(\sum \theta_i)$   $\nearrow$   $\frac{1}{m^2}$  not  $\frac{1}{m}$   $\nearrow$  Std.  $\sim \frac{1}{m}$  is Monte Carlo rule-of-thumb  
 $= \frac{1}{m^2} \sum Var(\theta_i)$  by independence  
 $= \frac{1}{m} \sigma^2$

So via Chebyshev,  $P[|\hat{L}_S(h) - L_D(h)| > \epsilon] \leq \frac{\sigma^2}{m \epsilon^2} \xrightarrow{m \rightarrow \infty} 0$  good!

That's for event  $E_h$ , so now union bound

$D^m(\epsilon) \leq \sum_{h \in H} D^m(E_h) \leq |H| \cdot \frac{\sigma^2}{m \epsilon^2} =: \delta$

So our sample complexity  $m_H^{uc}$  would be

$m_H^{uc}(\epsilon, \delta) = \frac{1}{\epsilon^2} \cdot \frac{1}{\delta} \cdot \sigma^2 \cdot |H|$   
 $\nearrow$  **vague but acceptable**  
 $\nearrow$  **not great but workable**  
 $\nearrow$  **not good**  
 $\nearrow$  **terrible:**  
 we want/need  
 a  $\log(|H|)$  or  
 similar

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Ex:  $\mathcal{H} = \{ \text{1D model, parameter is double precision number} \}$   
 so  $|\mathcal{H}| = 2^{64}$   
 then  $|\mathcal{H}|$  vs  $\log(|\mathcal{H}|)$  is a BIG DEAL

Union bound + Chebyshev = Bad idea

4) Chernoff-style bounds: Hoeffding, Chernoff, Bernstein etc  
 "Concentration Ineq." Exploit  $\sum$  iid structure

Many variants: see Handout.

We'll do Hoeffding

see also "Laplace transform methods"

lecture 16.3 Joel Tropp CMS/ACM 117 notes

First, some tools

Def A Rademacher (or symmetric Bernoulli) r.v.  $X$   
 means  $X = \begin{cases} 1 & \text{w.p. } 1/2 \\ -1 & \text{w.p. } 1/2 \end{cases}$   
 usually  $\{0,1\}$  output

and a Rademacher vector means each component is an independent Rademacher

Lemma "Exponentiated Markov"

( $X < 0$  ok but  $\lambda \geq 0$ ) If  $X$  is a r.v., and  $\lambda \in \mathbb{R}$  a constant,  $\lambda \geq 0$   
 $\mathbb{P}[X \geq \epsilon] \leq \exp(-\lambda \epsilon) \cdot \mathbb{E}[e^{\lambda X}]$  aka Moment Generating Funct MGF( $\lambda$ )  
 aka Laplace Transform. Bernstein 1927  
 Def: MGF  $M_X(\lambda) = \mathbb{E}[e^{\lambda X}]$   $\forall \lambda \in \mathbb{R}$   
 Def: CGF is  $\log(\text{MGF})$ , and additive for independent r.v.  
 proof  $\mathbb{P}[X \geq \epsilon] = \mathbb{P}[e^{\lambda X} \geq e^{\lambda \epsilon}]$  since  $X \mapsto e^{\lambda X}$  is monotonically increasing ( $\lambda \geq 0$ )  
 $\leq e^{-\lambda \epsilon} \mathbb{E}[e^{\lambda X}]$  via Markov's ( $e^m \geq 0$ )  $\square$

Lemma "Hoeffding's Lemma": Rademacher is sub-Gaussian

If  $X$  is Rademacher,  $\mathbb{E}[e^{\lambda X}] \leq e^{\lambda^2/2}$

Def: If  $\mathbb{E}[X] = 0$ , we call  $X$  "sub-Gaussian"  
 if  $\exists c > 0$  st.  $\forall \lambda \in \mathbb{R}$ ,  $\mathbb{E}[e^{\lambda X}] \leq e^{c\lambda^2}$   
 MGF

proof  $\mathbb{E}[e^{\lambda X}] = \frac{1}{2}e^{\lambda} + \frac{1}{2}e^{-\lambda} = \cosh(\lambda) \leq e^{\lambda^2/2}$

via Taylor Remainder Thm. or similar.  $\square$

Any variable w/ bdd. support is sub-Gaussian

now...

Thm (Hoeffding, simple, one-sided)

Let  $X_i$  be iid Rademacher,  $\vec{a} \in \mathbb{R}^n$ , then  $\forall \epsilon > 0$ ,

$\mathbb{P}\left[\sum_{i=1}^n a_i X_i \geq \epsilon\right] \leq \exp\left(\frac{-\epsilon^2}{2 \|\vec{a}\|_2^2}\right)$  Common cases:  
 $a_i = 1 \Rightarrow \|\vec{a}\|_2^2 = n$   
 $a_i = 1/n \Rightarrow \|\vec{a}\|_2^2 = 1/n$

Thm (Hoeffding, more general) See Vershynin Thm 2.6.2 for sub-Gaussian version

Let  $X_i$  be independent and bounded  $X_i \in [m_i, M_i]$ , then  $\forall \epsilon > 0$ ,

$\mathbb{P}\left[\left|\sum_{i=1}^n (X_i - \mathbb{E}[X_i])\right| \geq \epsilon\right] \leq 2 \cdot \exp\left(\frac{-2\epsilon^2}{\sum_{i=1}^n (M_i - m_i)^2}\right)$

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> proof:

Let  $\lambda > 0$  (TBD). wlog let  $\|\vec{a}\|_2 = 1$

By exponentiated Markov,

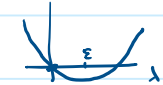
$$\begin{aligned} \mathbb{P}[\sum_i a_i X_i > \epsilon] &\leq \exp(-\lambda \epsilon) \mathbb{E}[\exp(\lambda \sum_i a_i X_i)] \quad \text{MGF} \\ &= \exp(-\lambda \epsilon) \mathbb{E}[\prod_i \exp(\lambda a_i X_i)] \quad \text{how exp works} \\ &= \exp(-\lambda \epsilon) \prod_i \mathbb{E}[\exp(\lambda a_i X_i)] \quad \text{independence individual MGFs} \\ &\leq \exp(-\lambda \epsilon) \prod_i \exp(\lambda^2 a_i^2 / 2) \quad \text{Hoeffding lemma} \\ &= \exp(-\lambda \epsilon) \exp(\lambda^2 / 2 \sum_i a_i^2) \quad \text{how exp works} \\ &= \exp(-\lambda \epsilon + \lambda^2 / 2) \quad \text{since } \|\vec{a}\|_2^2 = 1 \end{aligned}$$

now choose  $\lambda$  (to minimize the bound) — polynomial

i.e.,  $\lambda = \epsilon$ , so ...

$$\dots = \exp(-\epsilon^2 / 2) \quad \square$$

min  $-\lambda \epsilon + \lambda^2 / 2$



Set  $\frac{d}{d\lambda} = 0$ , Solve  $-\epsilon + \lambda = 0$  so  $\lambda = \epsilon$

Ex  $X_i \stackrel{iid}{\sim} \theta_i$

$X_i \stackrel{iid}{\sim} \text{Rademacher}$ ,  $i \in [m]$ , so  $\mathbb{E}[X_i] = 0$   
 $\mathbb{E}[X_i^2] = 1$  hence  $\sigma^2 = 1$

1) Hoeffding  $\mathbb{P}[\left|\frac{1}{m} \sum_{i=1}^m X_i\right| \geq \epsilon] \leq 2 \exp(-m \epsilon^2 / 2)$  i.e.  $\vec{a} \in \mathbb{R}^m$ ,  $a_i = \frac{1}{m}$ ,  $\|a_i\|_2^2 = \frac{1}{m}$

2) Chebyshev  $\mathbb{P}[\left|\frac{1}{m} \sum_{i=1}^m X_i\right| \geq \epsilon] \leq \frac{\sigma^2}{m \epsilon^2} = \frac{1}{m \epsilon^2}$  since  $\text{Var}(\frac{1}{m} \sum_{i=1}^m X_i) = \sigma^2 / m$

3) Exact: by symmetry,  $\mathbb{P}[\left|\frac{1}{m} \sum_{i=1}^m X_i\right| \geq \epsilon] = 2 \cdot \mathbb{P}[\frac{1}{m} \sum_{i=1}^m X_i \geq \epsilon]$

transform:  $X_i = \begin{cases} -1 \\ 1 \end{cases} \Rightarrow \tilde{X}_i = \begin{cases} 0 \\ 1 \end{cases}$  so  $\tilde{X}_i = \frac{1}{2}(X_i + 1)$   
 $X_i = 2\tilde{X}_i - 1$

$$\text{So } = 2 \cdot \mathbb{P}[\sum_{i=1}^m (2\tilde{X}_i - 1) \geq m \epsilon]$$

$$= 2 \cdot \mathbb{P}[\sum_{i=1}^m \tilde{X}_i \geq \frac{m}{2}(1 + \epsilon)]$$

$$= 2 \cdot (1 - \mathbb{P}[\sum_{i=1}^m \tilde{X}_i < \frac{m}{2}(1 + \epsilon)])$$

$$= 2(1 - F(\frac{m}{2}(1 + \epsilon))) \quad \text{where } F(x) \text{ is CDF of } n(\frac{1}{2}, m) \text{ binomial r.v.}$$

Ex

$m = 2000$ ,  $\epsilon = 0.1$

True  
 $6.8 \cdot 10^{-6}$

Hoeffding  
 $9.1 \cdot 10^{-5}$

Chebyshev  
 $.05$

$m = 109,000$   $\epsilon = 0.02$

$2.5 \cdot 10^{-10}$

$4.1 \cdot 10^{-9}$

$.025$

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... going back to our uniform convergence analysis ...

So via ~~Chernoff~~ <sup>Hoeffding</sup>, <sup>don't need Rademacher!</sup>  $\mathbb{P}[|\hat{L}_S(h) - L_D(h)| > \varepsilon] \leq \frac{\sigma^2}{m \varepsilon^2} \cdot 2 \cdot \exp(-m \varepsilon^2 / 2)$

That's for event  $E_h$ , so now union bound

$$D^m(E) \leq \sum_{h \in \mathcal{H}} D^m(E_h) \leq |\mathcal{H}| \cdot 2 \cdot \exp(-m \varepsilon^2 / 2) =: \delta$$

i.e., solving for  $m$ ,  $-m \varepsilon^2 / 2 = \log(\delta / 2|\mathcal{H}|)$

$$m = \frac{2}{\varepsilon^2} \log\left(\frac{2|\mathcal{H}|}{\delta}\right)$$

If " $x_i$ "  $\in [m_i, M_i]$  use generalized Hoeffding

Summary:

Corollary 4.6 U.C. for finite classes (binary classification, agnostic)

Let  $|\mathcal{H}| < \infty$ ,  $\ell$  be the 0-1 loss function, then  $\mathcal{H}$

has the uniform convergence property with

$$m_{\mathcal{H}}^{uc}(\varepsilon, \delta) \leq \left\lceil \frac{1}{\varepsilon^2} \log\left(\frac{2|\mathcal{H}|}{\delta}\right) \right\rceil$$

... and hence agnostic PAC learnable with

$$m_{\mathcal{H}}(\varepsilon, \delta) \leq m_{\mathcal{H}}^{uc}(\varepsilon/2, \delta)$$

factor of 2 discrepancy

my derivation above  
differs by a factor  
of 2, but not a  
big deal  
(don't trust me...  
but don't trust  
book either)